

SYSTEM://VISHVA — Portfolio Architecture Requirements Document

Oct 4, 2026 · @Vishva

1. Purpose and locked decisions

The portfolio is a static-first Astro site where every page ships as HTML and only four islands run JavaScript. Its single job: make a reviewer think "this person thinks in systems" within 20 seconds.

Identity statement (every decision is tested against it): AI systems engineering with security and reliability as first-class constraints.

The two source plans disagree on the stack (plan 1 says Next.js + Framer Motion + R3F; plan 2 says Astro + GSAP + plain Three.js). This ARD resolves it in favour of plan 2.

ID	Decision	Rejected alternative	Why
D-01	Astro + TypeScript, static output	Next.js App Router	Content site; React runtime over every page wastes the 150 KB budget
D-02	Plain Three.js in a TS module, no R3F	React Three Fiber	One canvas does not justify React + R3F weight
D-03	CSS + Web Animations API for ~70% of motion	Framer Motion everywhere	Browser-native, zero bundle cost
D-04	GSAP + ScrollTrigger, lazy-loaded, case-study pages only	GSAP site-wide	Only scroll storytelling needs it
D-05	SVG for every diagram a visitor must read	WebGL diagrams	Readable, selectable, accessible, animatable via stroke-dashoffset
D-06	Exactly one WebGL canvas, homepage only	Per-section canvases	Mobile GPU and battery
D-07	Astro View Transitions (ClientRouter) for page moves	Custom SPA router	Native API with built-in fallback

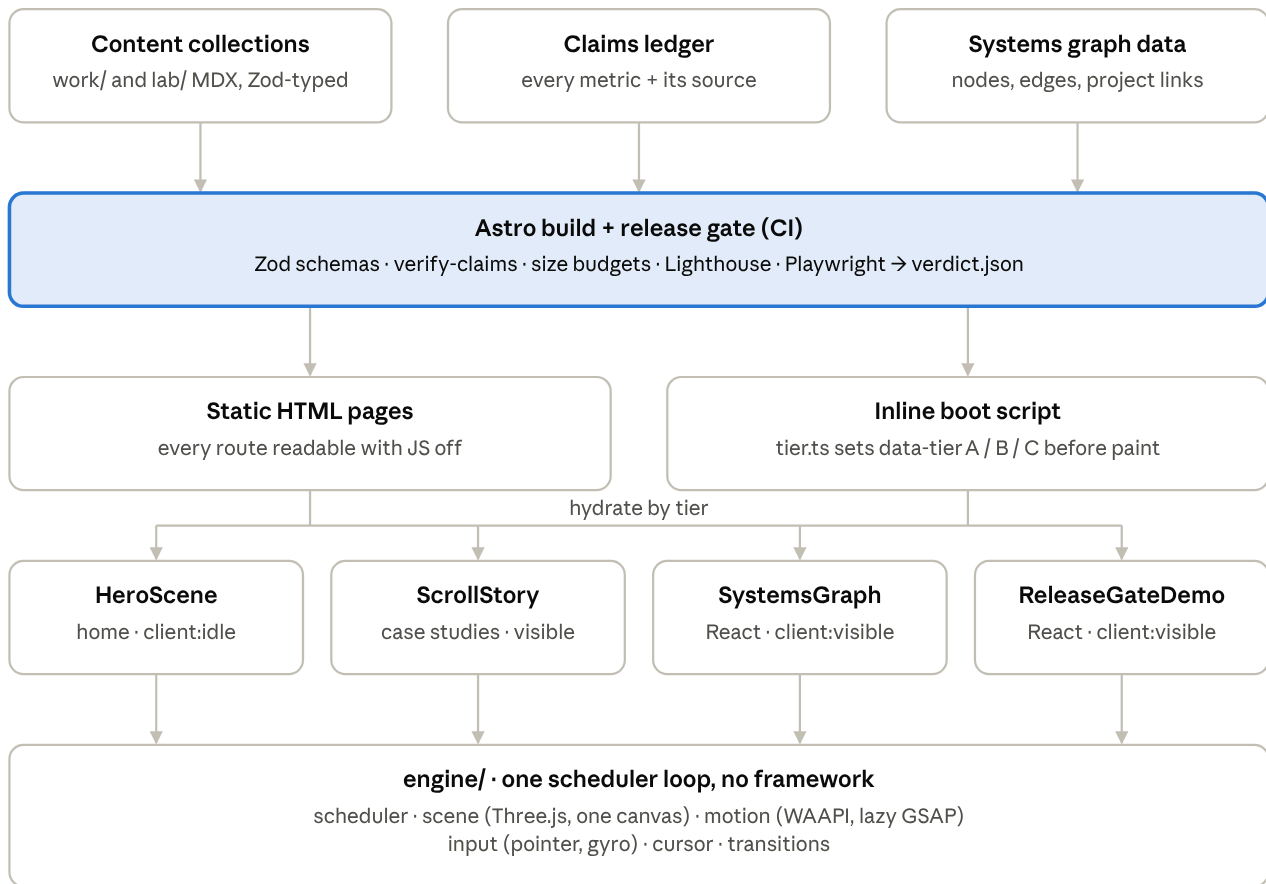
ID	Decision	Rejected alternative	Why
D-08	React islands only where state is real (Systems graph, QualityMesh demo)	React for everything	Islands architecture
D-09	No Lenis smooth scroll in V1	Lenis	Conflicts with plan 1's own "no scroll hijacking" rule (see §6)
D-10	Every metric on the site comes from one claims ledger with an evidence link	Hard-coded numbers in copy	"Measure before claiming" applied to the portfolio itself
D-11	Deploy to Vercel as static output	SSR	Nothing on the site needs a server in V1

Write each decision as a short ADR file in `docs/adr/` so the repo itself shows the reasoning.

2. System flow

Data moves strictly downward: content is validated at build, the release gate decides if the build ships, pages arrive as plain HTML, and only then does the boot script decide which islands wake up.

Facts flow one way: content, gated build, static HTML, four islands



system flow · content to engine, 5 layers

The gate's `verdict.json` is also read back by the footer status bar, so the site reports its own build health.

3. Folder structure

The repo splits into four layers that never import upward: `content` (facts) → `pages` (routes) → `components` (static HTML) → `engine` (runtime JS). Create these folders first, then fill them phase by phase.

```

system-vishva/
├─ astro.config.mjs          # static output, view transitions, sitemap
├─ package.json
├─ tsconfig.json            # strict, path aliases @engine @content @ui
├─ .size-limit.json         # JS budget per route
├─ lighthousec.json        # LCP / CLS / TBT gates

```

```

└─ .github/workflows/
  └─ release-gate.yml          # build → claims → budgets → e2e → deploy
└─ docs/
  └─ ARD.md                   # export of this document
  └─ adr/                     # 0001-astro-over-next.md ... one per decision
    └─ storyboards/          # rexi.md, maltrace.md ... scroll beat sheets
└─ public/
  └─ fonts/                   # self-hosted, subsetting woff2
  └─ og/                      # 1200×630 per page, generated at build
    └─ favicon.svg
└─ scripts/
  └─ verify-claims.ts         # fails build if a metric has no evidence
  └─ gen-og.ts                # OG images from page frontmatter
  └─ build-manifest.ts        # writes build hash + commit for status bar
└─ src/
  └─ content.config.ts        # Zod schemas for all collections
  └─ content/
    └─ work/                  # rexi.mdx maltrace.mdx qualitymesh.mdx
cloudshield.mdx
  └─ lab/                     # rexa.md access-risk-analyzer.md fulfillx.md
...
  └─ claims/                  # claims.yaml – the evidence ledger
  └─ principles.yaml          # "Model proposes. Software authorizes." ...
    └─ systems-graph.yaml     # nodes, edges, node → project links
  └─ pages/
    └─ index.astro
    └─ work/index.astro
    └─ work/[slug].astro
    └─ lab/index.astro
    └─ systems.astro
    └─ about.astro
    └─ contact.astro
    └─ plain.astro            # recruiter mode (see §7)
    └─ logs.astro             # changelog of the site itself
    └─ 404.astro
  └─ layouts/
    └─ Base.astro             # head, fonts, tier boot script, nav, status

```

```

bar
|   |   └─ CaseStudy.astro      # hero, storyboard slots, evidence drawer
|   |
|   └─ components/
|   └─ ui/                      # Panel Mono Metric StatusBar Nav
SystemRecord
|   |   └─ diagrams/           # ArchDiagram.astro Node.astro Edge.astro
|   |   └─ case/              # per-project static sections
|   |   └─ evidence/          # EvidenceDrawer.astro ClaimRef.astro
|   |
|   └─ islands/               # the ONLY hydrated code
|   |   └─ HeroScene.astro     # mounts engine/scene, client:idle
|   |   └─ ScrollStory.astro   # mounts engine/scroll, client:visible
|   |   └─ SystemsGraph.tsx    # React, client:visible
|   |   └─ ReleaseGateDemo.tsx # React, client:visible
|   |
|   └─ engine/                # framework-free runtime
|   |   └─ tier.ts              # capability detection → A / B / C
|   |   └─ scheduler.ts        # the one frame loop (see §5)
|   |   └─ input/
|   |   |   └─ pointer.ts
|   |   |   └─ gyro.ts         # iOS permission flow inside
|   |   |   └─ visibility.ts    # IntersectionObserver + page visibility
|   |   └─ scene/
|   |   |   └─ SystemScene.ts   # Three.js setup, DPR clamp, dispose()
|   |   |   └─ graph.ts         # node/edge layout from systems-graph.yaml
|   |   |   └─ shaders/        # node.vert node.frag edge.frag
|   |   └─ motion/
|   |   |   └─ tokens.ts        # durations, easings – mirrors tokens.css
|   |   |   └─ reveal.ts       # WAAPI mask reveals, line tracing
|   |   |   └─ count.ts        # metric count-up
|   |   |   └─ scroll.ts        # GSAP lazy import + ScrollTrigger setup
|   |   └─ transitions/        # per-project signature exits (see §7)
|   |   └─ cursor.ts           # context cursor, pointer:fine only
|   |
|   └─ styles/
|   |   └─ tokens.css           # colour, space, radius, z-index
|   |   └─ type.css            # two families, fluid scale
|   |   └─ motion.css          # keyframes + reduced-motion overrides
|   |   └─ global.css
└─ tests/

```

```

├─ e2e/
motion
├─ visual/
└─ engine/
# Playwright: nav, transitions, reduced
# screenshot diffs per tier
# Vitest: tier.ts, scheduler.ts, gyro math

```

4. Key files and what to build in each

Twelve files carry the architecture; get these right and the rest is content. Ordered by when you touch them.

File	Owns	Work on first	Done when
<code>styles/tokens.css</code>	Colours, radius 0/2/4px, spacing, z-index	The 8 colour tokens from plan 1§18, semantic names only (<code>-c-system</code> , <code>--c-risk</code> , <code>--c-info</code>)	No hex value appears anywhere else in the repo
<code>styles/type.css</code>	Grotesk + mono pairing, fluid scale	Pick one pair (suggest Inter Tight + JetBrains Mono), subset to Latin	Hero name renders with no layout shift (CLS 0)
<code>motion/tokens.ts</code>	4 durations, 3 easings	Mirror CSS custom properties so CSS and JS share one clock	Every animation imports from here
<code>content.config.ts</code>	Zod schemas for work, lab, claims	<code>work</code> schema: slug, title, thesis, status, stack, claimIds[], signature	Build fails on a missing field
<code>content/claims/claims.yaml</code>	Every number on the site	List each metric (323+ tests, 50/50 pricing parity, 71 golden evals, 94,958 API calls, 54 features, 95%) with source URL	<code>verify-claims.ts</code> passes
<code>engine/tier.ts</code>	Decides A / B / C once at boot	Read reduced-motion, WebGL2 support, <code>hardwareConcurrency</code> , <code>deviceMemory</code> , <code>save-data</code> ; set <code>data-tier</code> on <code><html></code>	Unit tests cover all 3 outcomes
<code>engine/scheduler.ts</code>	The single <code>requestAnimationFrame</code> loop	Subscribe / unsubscribe API, auto-sleep when no subscriber is dirty	Idle page shows 0 frames in DevTools Performance
<code>engine/scene/SystemScene.ts</code>	Three.js renderer, camera, graph mesh	DPR clamp (A: 1.5, B: 1.25), instanced nodes, <code>dispose()</code> on navigation	50+ FPS on a mid-range Android
<code>engine/input/gyro.ts</code>	Orientation → normalised x/y, lerped	iOS <code>requestPermission()</code> from a tap, clamp to 3-8 px	Works on iPhone Safari after one tap; silent elsewhere
<code>layouts/Base.astro</code>	Head, inline tier script, nav, status bar	Inline tier script before first paint so CSS can branch on <code>data-tier</code>	No flash of the wrong tier

File	Owns	Work on first	Done when
<code>components/diagrams/ArchDiagram.astro</code>	SVG architecture from a node list	Takes <code>nodes[]</code> + <code>edges[]</code> props; outputs paths with <code>pathLength="1"</code> for easy tracing	Rexi pipeline renders static with zero JS
<code>pages/work/[slug].astro</code>	Case-study route	<code>getStaticPaths</code> from the work collection, <code>CaseStudy</code> layout	All four projects build from MDX alone

5. Runtime loop inventory

The site has exactly one frame loop (`scheduler.ts`); every other row below is an event source feeding it. Anything else that calls `requestAnimationFrame` directly is a bug. When no subscriber is dirty, the loop stops.

Loop / listener	Runs on	Starts when	Stops when	Tier C
Scheduler rAF	Every page with an island	A subscriber marks itself dirty	No dirty subscriber for 1 frame	Never starts
Hero render	Home	Hero visible AND (pointer moved OR gyro changed OR idle drift active)	Hero off-screen, tab hidden, or drift budget spent	Static SVG grid instead
Idle drift	Home	Page load	After 6 s with no input, then sleeps until input	Off
Pointer listener	Home, Systems	<code>pointer: fine</code> and island mounted	Island unmounted	Passive, no render
Gyro listener	Home, mobile	User grants permission (iOS) or API present (Android)	Hero off-screen or tab hidden	Off

Loop / listener	Runs on	Starts when	Stops when	Tier C
ScrollTrigger	Case studies	Story section enters viewport (<code>client:visible</code>)	Page navigation → <code>ScrollTrigger.killAll()</code>	Sections render final state
Cursor follower	All pages, desktop	<code>pointer: fine</code> and tier A	Pointer leaves window	Native cursor
Count-up	Metric blocks	Block enters viewport	Count finishes (≤ 800 ms)	Final number shown

Cleanup rule: every island exports `mount()` and `destroy()`. Astro fires `astro:before-swap` on navigation; `destroy()` must cancel listeners, kill ScrollTriggers, and call `renderer.dispose()` plus geometry/material disposal. A leaked WebGL context after 10 navigations is a release blocker.

6. Loopholes in the two plans

Twelve gaps would break the site or contradict its own principles if built as written. Each has a fix already folded into this ARD.

#	Loophole	Risk	Resolution
L-01	Plan 1 picks Next + Framer + R3F; plan 2 picks Astro + GSAP + Three	Two half-built stacks	D-01, D-02: Astro wins
L-02	"Subtle animated connections" vs "no continuous render when idle"	Either dead hero or battery drain	Idle drift for 6 s, then sleep until input (§5)

#	Loophole	Risk	Resolution
L-03	Three.js core is roughly 150 KB gzipped on its own vs a 150 KB initial-JS budget	Budget blown on page one	Three loads <code>client: idle</code> after LCP; budget applies to pre-LCP JS; static SVG grid paints first
L-04	iOS Safari blocks device orientation until <code>DeviceOrientationEvent.requestPermission()</code> is called from a tap	Gyro silently never works on iPhone	Small "ENABLE DEPTH" chip on iOS only; Android just works
L-05	"Fake command lines" banned, yet "command execution" animation proposed	Site violates its own rule	Every terminal sequence replays a real captured log stored in <code>content/</code> with its source
L-06	Metrics (323+ tests, 95%, 94,958 calls) hard-coded in copy	Stale or unverifiable claims in interviews	Claims ledger + <code>verify-claims.ts</code> gate (D-10)
L-07	"Every project gets its own motion identity" vs "one consistent motion system"	Visual chaos	Shared primitives (trace, mask, count, node) + one signature exit per project (§7)
L-08	Lenis suggested while "custom scroll hijacking" is banned	Self-contradiction, jank on mobile	No Lenis in V1 (D-09)

#	Loophole	Risk	Resolution
L-09	Custom system cursor has no touch or keyboard story	Lost affordance on phones, a11y fail	Cursor only on pointer: fine + tier A; all labels also exist as visible text and focus styles
L-10	Scroll-pinned GSAP sections hide content from screen readers and keyboard users	Inaccessible case studies	Every storyboard step is real HTML in reading order; GSAP only animates it
L-11	Lab statuses ("ACTIVE") go stale silently	Abandoned work looks dishonest	lastTouched date required in schema; status auto-shows "DORMANT" after 90 days
L-12	A 30-second recruiter skim was never designed for	Great site, lost interview	/plain recruiter mode + one-page PDF link in nav (§7)

Open question: CloudShield and FulfillX have no source facts yet; their pages stay out of Phase 1 until claims exist for them.

7. Creative work

The strongest creative move is making the portfolio obey the same rules as the projects it shows: the site is itself a gated, measured, fail-closed system. Ten ideas, each tied to the identity.

1. **The site ships through its own QualityMesh.** The CI release gate (claims verified, budgets met, e2e green) writes a JSON verdict at build time. The footer status bar reads it: `BUILD 7f3a1c · GATE: PASS · LCP 1.9s · JS 84KB`. The QualityMesh case study links to this live gate as its demo.
2. **Evidence drawer.** Any metric with a dotted underline opens a side drawer showing its source: test output, commit, eval file. "Measure before claiming" becomes something a visitor can click.
3. **Model proposes, software authorizes — as an interaction.** On the Rexi page, a visitor types an order sentence; a canned interpreter output shows the proposed intent in muted text, then the deterministic layer stamps it `VALIDATED` or `REJECTED` in system green or risk red. All outputs are pre-recorded from real runs (no live LLM, no fake).
4. **Signature exits.** One per project, built from shared primitives: Rexi's record collapses into a waveform that becomes the pipeline; MalTrace's row resolves into a SHA-256 that hashes character by character into the case study; CloudShield streams audit lines across the wipe; QualityMesh stamps `PASS` on the outgoing page.
5. **Graph that knows the visitor's path.** Hero nodes for pages already visited stay lit on return (sessionStorage only). The map fills in as they explore.
6. **/plain recruiter mode.** Same content, zero motion, single column, print-ready, linked as `[PLAIN]` in the nav. Shows confidence: the work survives without the effects.
7. **/logs as a changelog.** The site's own git history rendered as a telemetry feed. Proves the portfolio is maintained, not abandoned.
8. **404 as a fail-closed event.** `REQUEST DENIED · ROUTE NOT IN ALLOWLIST · RETURNING TO KNOWN STATE` with a link home. On-theme, one line, no gimmick.
9. **Tier is visible.** Status bar shows `RENDER: TIER B` on phones. Turns a performance decision into a talking point.
10. **Keyboard command bar.** `/` opens a minimal command palette (`open rexi`, `goto lab`, `mode plain`). Real navigation, not a fake terminal.

Cut list if time runs short: 5, 9, 10. Never cut 1, 2, 6.

8. Budgets and release gate

A build that misses any hard budget does not deploy. The gate runs in `.github/workflows/release-gate.yml` and its verdict feeds the footer status bar (§7, idea 1).

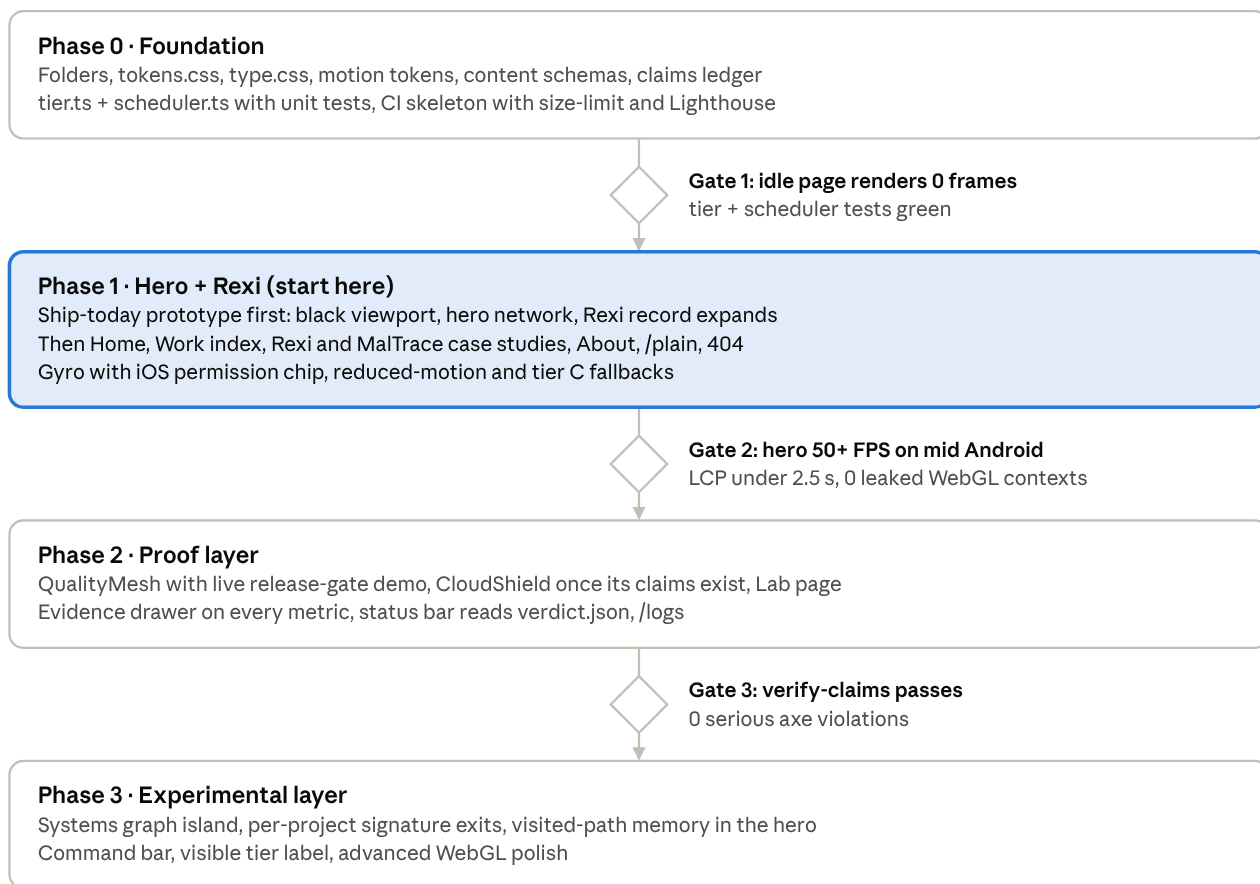
Gate	Threshold	Tool	Hard / soft
Pre-LCP JavaScript	< 60 KB gzipped (Three.js excluded, loads after)	size-limit	Hard
Total JS, home	< 250 KB gzipped incl. Three.js	size-limit	Hard
Total JS, case study	< 90 KB gzipped incl. GSAP	size-limit	Hard
LCP, mobile throttled	< 2.5 s	Lighthouse CI	Hard
CLS	< 0.02	Lighthouse CI	Hard
Hero FPS, mid Android	≥ 50 tier B, ≥ 30 tier C-fallback	Manual + Playwright trace	Soft
Idle frames, home	0 after 6 s no input	Playwright performance trace	Hard
WebGL contexts	1, and 0 leaked after 10 navigations	Playwright e2e	Hard
Claims	Every metric has a resolvable source	<code>verify-claims.ts</code>	Hard
Reduced motion	No transform animation runs	Playwright with <code>reducedMotion: 'reduce'</code>	Hard
Accessibility	0 serious axe violations	axe-core in Playwright	Hard
Fonts	≤ 2 families, ≤ 4 files, subsetting woff2	build script	Soft

Test devices: one mid-range Android on Chrome, one iPhone on Safari, desktop Chrome and Firefox.

9. Build order

Start with Phase 0 today, then build the hero plus one Rexi transition before any other page; nothing advances past a gate that fails.

Prove the hero and one Rexi transition before building the rest



build order · 4 phases, 3 gates

Today's checklist

- ☐ `npm create astro@latest system-vishva` (empty template, strict TS)
- ☐ Create every folder in §3, with an empty `index` or `.gitkeep`
- ☐ Write ADRs 0001-0011 from the decisions table in §1
- ☐ Fill `tokens.css` and `type.css`; self-host the two font families
- ☐ Write `tier.ts` and `scheduler.ts` with Vitest tests
- ☐ Start `claims.yaml` with every number from the Rexi and MalTrace plans, each with a source
- ☐ Write the Rexi storyboard in `docs/storyboards/rexi.md` (beats, scroll %, what animates, what it communicates)